

EP-38

**high speed
cassette system
eprom board**

users manual



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high speed cassette system eprom board

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INSTRUCTION MANUAL RELEASE 1

INTRODUCTION

You should read this manual COMPLETELY before installing or using your new EP-38.

The EP-38 EPROM Board is designed to work with the TC-8 High Speed Cassette System for the TRS-80 Model I. This board will fit within the existing TC-8 Cabinet and provide immediate access to the UTIL operating system. The EPROM, provided on the board, contains the latest release of the UTIL operating system and provides many new options and features for the TC-8 owner.

Since the new UTIL operating system is in EPROM, none of your RAM is used up by the operating system. Programs that require all of your RAM space can now be used with no interference by UTIL. All of your existing TC-8 software is compatible with the new UTIL operating system.

NEW FEATURES

The latest release of the UTIL operating system now honors the Model I Keyboard Soft Vector. This means that programs like lower case drivers will no longer present a problem to the TC-8 user. Refer to the section on IMPLEMENTATION for both lower case drivers and keyboard debounce.

UTIL now honors the Model I BASIC Program Start Soft Vector. Now you can perform functions like Merge and you can transfer BASIC files back and forth between programs like Level III BASIC and Disk BASIC. Refer to the section on IMPLEMENTATION for merge and inter-program transfers.

UTIL now supports BASIC data buffers of any length. The default length for data files remains at 128 but if you allocate some of your Model I RAM you can reduce the time (and tape) required for data transfers by half or more. If you allocate 384 Bytes of RAM you can reduce data times by over half and 1K buffers make a fantastic reduction in data file handling time for large data bases. Refer to the section on IMPLEMENTATION of data file buffers.

IMPLEMENTATION

KEYBOARD DEBOUNCE

A keyboard debounce routine is still provided in the EPROM version of UTIL. There are two entry points for UTIL - the first entry point (executed by: SYSTEM /12288) automatically invokes the keyboard debounce routine. If you don't want the debounce routine or you have a special keyboard driver (see the lower case driver section) then enter UTIL through the second entry point at /12290.

LOWER CASE DRIVERS

UTIL will honor any keyboard driver that has previously modified the Keyboard Soft Vector. In order to install such a driver, you must load that driver and execute it. This will substitute that driver's values into the Keyboard Soft Vector. Simply execute UTIL at /12290 and UTIL will now honor that program.

MERGE

Merge is really a misnomer for the function we are about to describe. It is properly called concatenate or append. This means to join together or to add on. The following procedure will allow you to join two BASIC programs together into one program.

Let us assume that you have two programs that you want to append. The first program "MASTER" contains the base line program and the second program, "SLAVE", contains either subroutines or data that is to be used by "MASTER". The following procedure will describe how to join these two programs into one.

The first requirement is that the line numbers of the two programs are mutually compatible. This means that all the line numbers in the program "SLAVE" must be greater than the line numbers in the program "MASTER". The programs could have been written that way or by using a program called "TWOHAF", published in 80 Microcomputing, July 1981, by Gil Spencer. You can renumber your programs to meet this requirement.

- (1) Load your first program "MASTER"
- (2) Type the following:
 ?PEEK(16548),PEEK(16549)
- (3) Write the values printed out on a piece of paper
- (4) Type the following:
 X=PEEK(16333)+PEEK(16334)*256-2
 Y=INT(X/256)
 Z=X-Y*256


```
POKE 16548,Z
POKE 16549,Y
(5) Load the second program "SLAVE"
(6) Type in the following:
    POKE 16548,***** Where ***** is the first value
        written down in step (3)

    POKE 16549,----- Where ----- is the second value
        written down in step (3)
```

Both programs will now be in memory together and they can be run or saved as one program as you desire.

INTER-PROGRAM TRANSFERS

This procedure can not cover all possible cases or problems involved in transferring programs between different BASICS but it will outline the methods required to perform this function.

Since UTIL now honors the BASIC Program Start Soft Vector, you can load any BASIC file previously saved through UTIL into any memory locations desired. All you need to do is POKE a new start address into the BASIC Program Soft Vector and perform a normal LOAD. Once the BASIC program is in memory, you can load your new BASIC (Level III or Disk) and poke the new program start and stop addresses into its Soft Vectors. The reverse procedure will also hold true. You should note that when you return to the "regular" BASIC, you should re-execute UTIL so that the normal SAVE, LOAD, and other UTIL functions will be performed through the TC-8. Refer to the KEY MEMORY ADDRESSES section for the BASIC soft vectors.

DATA FILE BUFFERS

UTIL automatically allocates space for 128 Byte buffers. This memory is out of the range of your BASIC programs. This means that none of your MODEL I BASIC program RAM is used by the TC-8. However, UTIL will now allow you to expand the size of your data file buffer beyond 128 Bytes. There are four memory locations in the RAM that can be changed to move the data file buffer to any location in memory and to make it any size you desire. You should be very careful when you take advantage of this feature, it could "BYTE" you. More on this later. The two memory locations 16505 and 16506 contain the length of the data buffer. If you were to print the PEEK of those values after executing UTIL you would find them to be 128 and 0. This is equivalent to 128. How you say? To make a long story short (the Model I - Z80 processor causes all this) take the second

value * 256 and add the first value and you get 128.

To create a 1-K buffer (1024 Bytes) type in the following:

```
POKE 16506,INT(1024/256)
POKE 16505,1024-INT(1024/256)*256
```

This will set the proper buffer length for a 1-K buffer. If you want any other buffer size (larger or smaller) just change the value of 1024 in the above example.

The memory location for the data buffer is calculated in the same way. The two memory locations 16507 and 16508 contain the address of the starting point of the data buffer in RAM. If you were to print the PEEK of those values after executing UTIL you would find them to be 96 and 66. This is equivalent to 16992 which is the EP-38's data file buffer address. The same procedure used to calculate the buffer length will be used here. Assume you wanted to put your 1-K buffer at memory location 31743 (your top 1-K in a 16-K Model I).

```
POKE 16508,INT(31743/256)
POKE 16507,31743-INT(31743/256)*256
```

You now have a 1-K buffer starting at RAM location 31743.

BE CAREFUL when you perform this procedure, a number of things could go wrong! The first thing to worry about is the memory reserved for the buffer. You must have pre-set MEMORY SIZE to at least 2 Bytes less than the start of your buffer or the buffer must be protected in some other way. In our example you should have set MEMORY SIZE to 31741 when you powered up your system. The second thing to worry about is do you really have RAM at the starting location specified and is there enough to hold the buffer size you specified? The last item to take into consideration is that in order to read these data files you must re-specify the buffer start address when you power up your system (or include this procedure in your program). This all appears very complicated at first but with good programming practice this feature can save you considerable time in programs with large data bases. One last item, if you want to share non standard length data files with a friend that doesn't have the new EP-38 and the new operating system, you will have to specify 128 byte buffers starting at memory 16992. You might also tell him how great the EP-38 is and to upgrade his system too!

To execute the EP-38 UTIL after installation simply go to SYSTEM and type /12288 (for keyboard debounce) or /12290 (for non keyboard debounce) and you're on your way.

KEY MEMORY ADDRESSES

DECIMAL	HEX	PURPOSE	
16406	4016	BASIC Keyboard Soft Vector	- Low BYTE
16407	4017		- High BYTE
16548	40A4	BASIC Program Start Vector	- Low BYTE
16549	40A5		- High BYTE
16633	40F9	BASIC Program End Vector	- Low BYTE
16634	40FA		- High BYTE
16505	4079	UTIL Data File Buffer length	- Low Value
16506	407A		- High Value
16507	407B	UTIL Data File Buffer Start	- Low Value
16508	407C		- High Value
12388	3000	EP-38 UTIL Entry Point (K.B. Debounce)	
12290	3002	UTIL Entry Point (L.C. Driver)	
12293	3005	EP-38 UTIL MOTON Subroutine Entry Point	
12296	3008	UTIL GETID Subroutine Entry Point	
12299	300B	UTIL GETID2 Subroutine Entry Point	
12302	300E	UTIL RFILE Subroutine Entry Point	
12305	3011	UTIL MOTOFF Subroutine Entry Point	
12308	3014	UTIL WRTID Subroutine Entry Point	
12311	3017	UTIL WFILE Subroutine Entry Point	
12314	301A	UTIL KBFIX Subroutine Entry Point	

WARRANTY

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The EP-38 has been factory assembled and tested before shipment. It requires careful handling during installation.

The EP-38 UTIL operating system has been extensively tested and there are no known bugs.

```
*****  
*                                     *  
*          CAUTION                   *  
*                                     *  
*****
```

The EP-38 contains circuitry that can be damaged due to static electricity or careless handling. When handling the EP-38 before it is installed into the TC-8 cabinet you should work, as much as possible, in a static free environment. Try not to walk on nylon rugs or pet your cat.

If the terms of the Warranty are not acceptable, you may return the EP-38 within 10 days of the shipping invoice date in its unopened sealed container for a full refund.

FIGURES

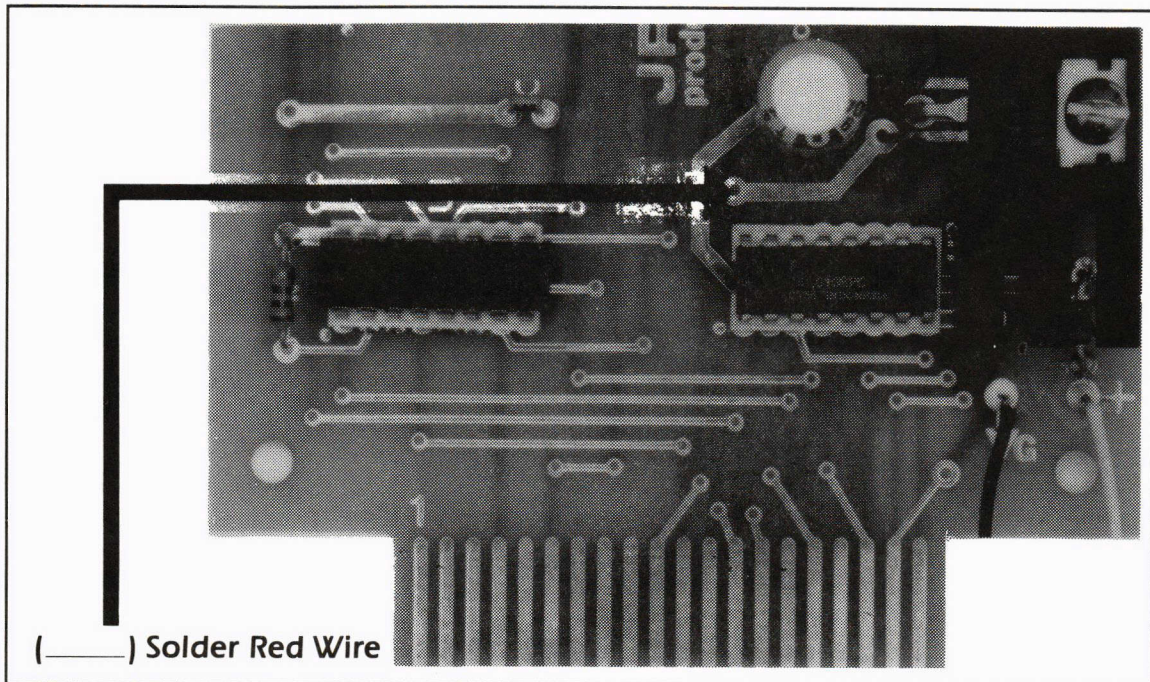


Figure 1

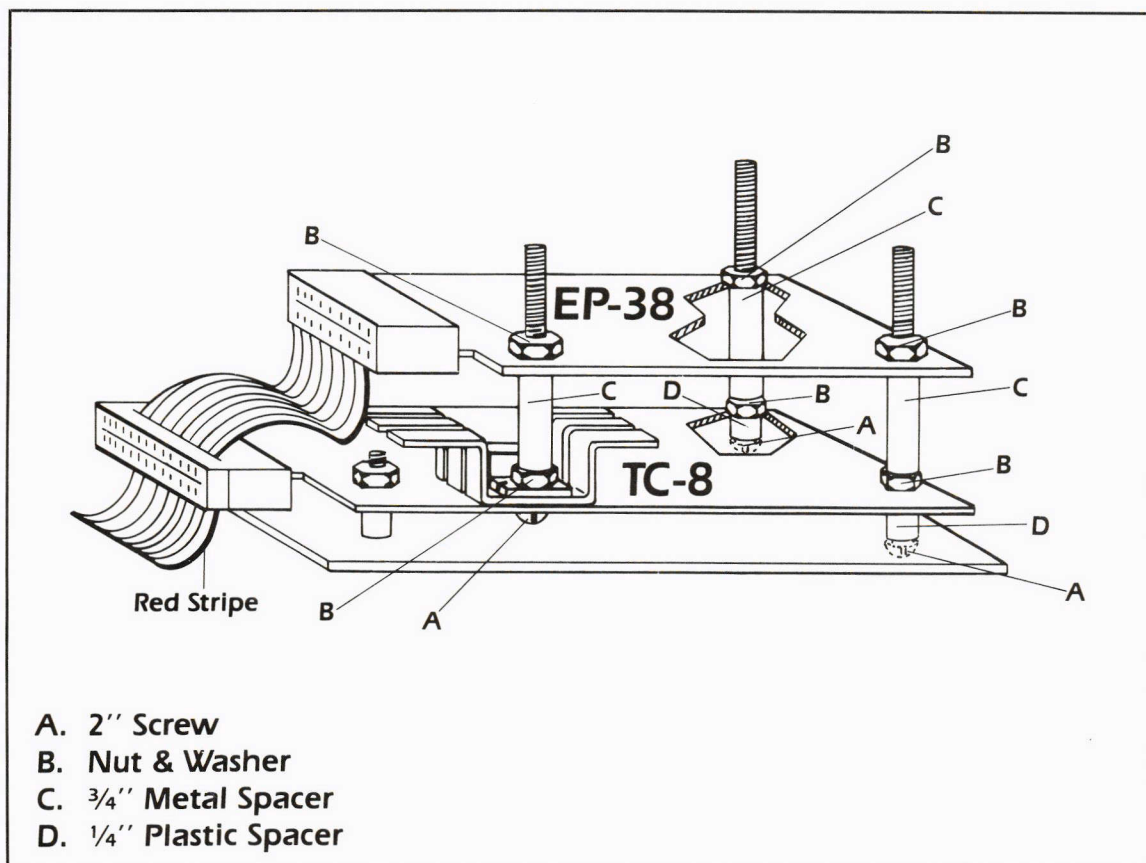


Figure 2

HARDWARE INSTALLATION

Do not proceed with this section unless you have read the complete manual including the Warranty.

PROCEDURE I - Disassembling your TC-8

In order to install the EP-38, you will have to remove the TC-8 from its cabinet. Remove all cables (power, ribbon, cassette) from the TC-8 and move it to a well lighted, static electricity free, work bench or table. Get a small bowl to hold the parts you are working with. You will also need a small Phillips and common screw driver, a small socket driver set or a pair of pliers, some masking or scotch tape and a soldering iron. Refer to the Assembly Instructions in your TC-8 Manual if there is any question about parts description or locations not covered here.

- (1) Remove the four sheet metal screws holding the white cover to the blue base. Use a small Phillips screw driver only. Carefully separate the white top from the blue bottom. There will be two wires connecting the top to the TC-8 circuit board.
- (2) Remove the four bolts (1/2 ") holding the TC-8 board on the blue bottom. Use a small common screw driver only.
- (3) Gently lift the TC-8 board out of the blue bottom. Put the bolts, nuts, washers and 1/4" spacers in the bowl.

PROCEDURE II - Replacing the Regulator Bolt

- (1) The regulator and heat sink on the TC-8 board is held on by a 1/4" bolt and nut. Remove this bolt now. Refer to Detail 6 in the TC-8 Assembly Instructions.
- (2) Remove a 2" bolt from the EP-38 parts package and install it from the bottom side of the TC-8 board through the regulator and heat sink holes. Firmly tighten down a washer and nut so that the regulator and heat sink are tight against the TC-8 board.

PROCEDURE III - Power Connection

At this time remove the EP-38 from its sealed package. You should keep in mind the warranty and the warnings about static electricity.

During this procedure you must be very careful not to drip any

solder on either the TC-8 or EP-38 circuit boards.

(1) Take the red wire from the EP-38 circuit board and solder it in the hole in the TC-8 Circuit Board as indicated in Figure 1 of this manual. Be careful that you select the proper hole and that you don't drip or splash any solder. Solder from the bottom side of the TC-8 board only.

PROCEDURE IV - Re-installing the TC-8 board

(1) Place two 1/2" bolts through the underside of the blue bottom on the "short end" - see Detail 10 in the TC-8 Assembly Instructions. These bolts were removed in Procedure I. Use masking tape to hold them in place.

(2) Place two 2" bolts from the EP-28 parts package through the underside of the blue bottom on the "long end". Use masking tape to hold them in place.

(3) Place the blue bottom resting on its rubber feet and slip the four 1/4" spacers over the bolts.

(4) Carefully install the TC-8 circuit board on the four bolts and tighten four washers and nuts to firmly hold the TC-8 board in its cabinet.

PROCEDURE V - Installation of the EP-38

(1) Slip the three 7/8" metal spacers from the EP-38 parts package over the three bolts sticking up out of the TC-8.

(2) Carefully install the EP-38 circuit board on the three bolts and tighten three washers and nuts from the EP-38 parts package to firmly hold the EP-38 board in the cabinet.

PROCEDURE VI - Ribbon cable connection

You must use extreme care in this procedure. The EP-38 only has a three point mount and as such it will not be perfectly rigid. Support the EP-38 board with one hand as you install the connector. You must use the new ribbon cable provided in your EP-28 package. It has three connectors on it. Two of them are about 2" apart and the third is at the far end of the cable.

(1) Take the end connector (near the double connector) and place it CAREFULLY on the EP-38. The red strip should be on the left as you face the connector.

(2) Place the inner most of the double connectors carefully on the TC-8. The red stripe should be on the left as you face

the connector.

PROCEDURE VII - Closing it up

It's time to close up your handy work and start using your new system.

(1) Do a final inspection of your work. No solder bridges but a good solder connection on the power connecton. All nuts and bolts tight.

(2) Carefully reinstall the white top in the blue bottom. Line up the four holes in each piece. Refer to Detail 12 in the TC-8 Assembly Manual.

(3) Screw in the four sheet metal screws using the Phillips screw driver.

(4) Attach the ribbon cable to the Model I.

(5) Attach the power connector from the power supply.

(6) Power up your Model I (no need to set any Memory Size for the new operating system).

(7) Go to SYSTEM and type /12288 you should see
UTIL 5.x READY

(8) Happy Programing

REWARD

JPC Products Company will exchange 5 of our C-10 Ferric Data Cassettes for the return of your old TC-8 Ribbon Cable. Just return it, well packaged, in the EP-38 shipping box. Be sure to include your name and address.

TECHNICAL ASSISTANCE

JPC Products Company will provide technical assistance by telephone between the hours of 5:00 to 6:00 PM Mountain Time.

COMMENTS & SUGGESTIONS

If you like our products and service tell your friends and the magazine editors. If you don't, tell us and if possible we'll fix it.

In order to use Radio Shacks Scripsit program with JPC Products FASTEX patches on the EPROM version of UTIL (Version 5.0 or later) you will have to perform the following procedure:

- (1) Load your version of SCRIPSIT that has the FASTEX patches.
- (2) Perform the following POKE's as commands (do not incorporate them into a BASIC program).

```
*****
* POKE ADDR * POKE VALUE * COMMENTS *
*****
*          *          *          *
* ( ) 28761 * 00      * UTIL ENTRY POINT *
* ( ) 28762 * 48      *          *
* ( ) 28788 * 00      * UTIL ENTRY POINT *
* ( ) 28789 * 48      *          *
* ( ) 28431 * 05      * MOTON ENTRY POINT *
* ( ) 28432 * 48      *          *
* ( ) 28445 * 08      * GETID ENTRY POINT *
* ( ) 28446 * 48      *          *
* ( ) 28455 * 11      * GETID2 ENTRY POINT *
* ( ) 28456 * 48      *          *
* ( ) 28458 * 14      * RFILE ENTRY POINT *
* ( ) 28459 * 48      *          *
* ( ) 28515 * 17      * MOTOFF ENTRY POINT *
* ( ) 28516 * 48      *          *
* ( ) 28521 * 17      * MOTOFF ENTRY POINT *
* ( ) 28522 * 48      *          *
* ( ) 28642 * 20      * WRITID ENTRY POINT *
* ( ) 28643 * 48      *          *
* ( ) 28645 * 23      * WFILE ENTRY POINT *
* ( ) 28646 * 48      *          *
*****
```

- (3) You should verify these POKES with ?PEEK's. If there are ANY errors, you should start over at step (1) of this procedure. The reason is that you may have POKE'd the wrong memory location and this would probably cause SCRIPSIT to act very strange (or not at all) at times.
- (4) Now make a new copy of SCRIPSIT following the original instruction in the FASTEX manual.

If you need to use the CONVRT program then you will have to patch it using the same procedure as used for the SCRIPSIT program.

- (1) Load your version of CONVRT that has the FASTEX patches.
- (2) Perform the following POKE's as commands (do not incorporate them into a BASIC program).

```
*****
*   POKE ADDR * POKE VALUE * COMMENTS                                     *
*****
*           *           *           *
* ( ) 28750 * 00         * UTIL ENTRY POINT                             *
* ( ) 28751 * 48         *                                           *
* ( ) 28777 * 00         * UTIL ENTRY POINT                             *
* ( ) 28778 * 48         *                                           *
* ( ) 28431 * 05         * MOTON ENTRY POINT                           *
* ( ) 28432 * 48         *                                           *
* ( ) 28445 * 08         * GETID ENTRY POINT                           *
* ( ) 28446 * 48         *                                           *
* ( ) 28455 * 11         * GETID2 ENTRY POINT                           *
* ( ) 28456 * 48         *                                           *
* ( ) 28458 * 14         * RFILE ENTRY POINT                           *
* ( ) 28459 * 48         *                                           *
* ( ) 28515 * 17         * MOTOFF ENTRY POINT                           *
* ( ) 28516 * 48         *                                           *
* ( ) 28521 * 17         * MOTOFF ENTRY POINT                           *
* ( ) 28522 * 48         *                                           *
* ( ) 28642 * 20         * WRITID ENTRY POINT                           *
* ( ) 28643 * 48         *                                           *
* ( ) 28645 * 23         * WFILE ENTRY POINT                           *
* ( ) 28646 * 48         *                                           *
*****
```

- (3) You should verify these POKES with ?PEEK's. If there are ANY errors, you should start over at step (1) of this procedure. The reason is that you may have POKE'd the wrong memory location and this would probably cause CONVRT to act very strange (or not at all) at times.
- (4) Now make a new copy of CONVRT following the original instruction in the FASTEX manual.

